SIMATS ENGINEERING
ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED
CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct
I S.Blessika(192525251) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: S.Blessika

Assessment Tasks
Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy.

Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

Introduction

A media company migrating to the cloud requires a highly scalable, reliable, and cost-effective architecture. The cloud enables the platform to handle millions of concurrent users while ensuring low latency and minimal buffering.

Cloud Architecture

- Store video files in **Cloud Object Storage**.
- Use a **Content Delivery Network (CDN)** to cache videos near users.
- Deploy **Load Balancers** to distribute incoming traffic.
- Run streaming servers in **Auto Scaling Groups**.
- Use **Redis/Memcached** for session caching.
- Store user information in a managed cloud database.
- Monitor the system using cloud monitoring services.

Load Balancing and Auto-Scaling

Load Balancing

- Distributes requests among multiple servers.
- Prevents server overload.
- Removes unhealthy servers automatically.
- Improves availability and reliability.

Auto-Scaling

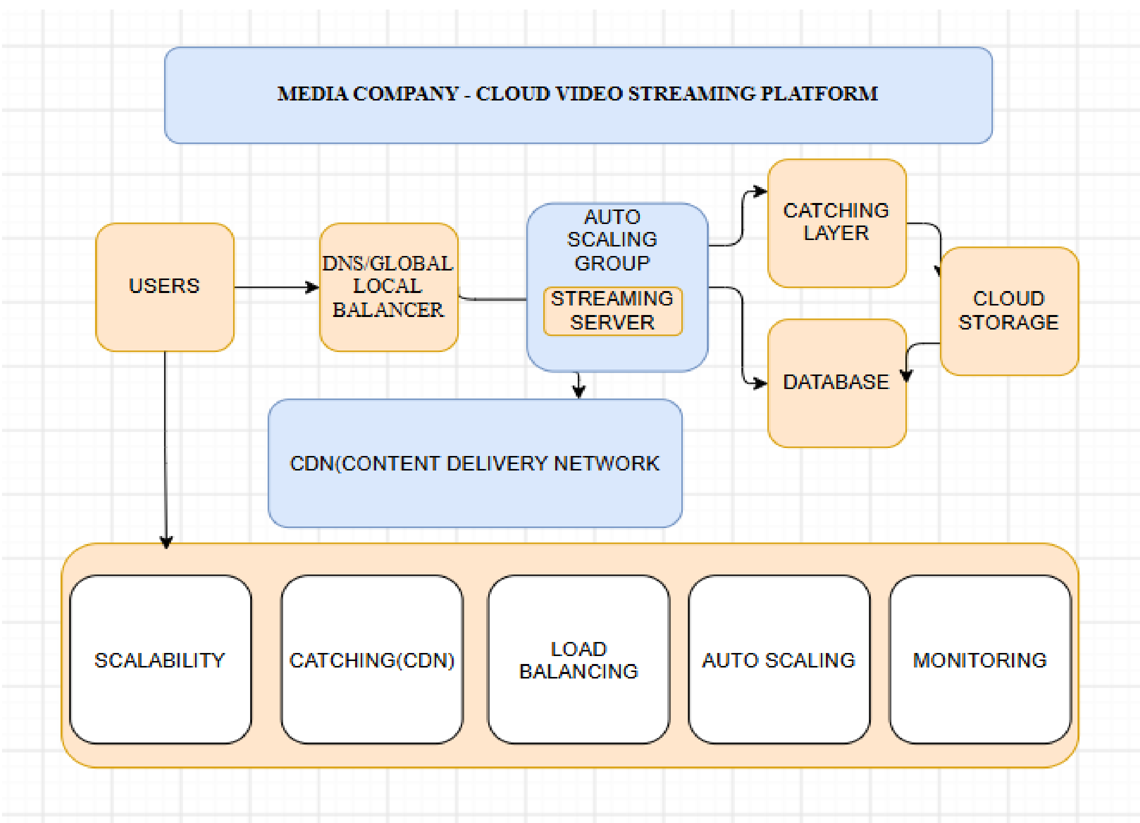
- Adds new servers during peak traffic.
- Removes idle servers during low traffic.
- Maintains application performance.
- Optimizes cloud resource usage.

Cost Optimization Strategies

- Use CDN caching to reduce bandwidth usage.
- Compress video files to save storage.
- Store old videos in archive storage.
- Use Reserved or Spot Instances.
- Enable Auto Scaling to reduce idle infrastructure costs.

Advantages

- High scalability
- Minimal buffering
- Better user experience
- High availability
- Reduced operational cost.



2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

Introduction

A fintech startup needs a consistent deployment strategy across multiple cloud providers. Containerization and orchestration simplify application deployment, scaling, and management.

Proposed Solution

- Package applications using **Docker** containers.
- Use **Kubernetes** for orchestration.
- Deploy clusters on AWS, Azure, and Google Cloud.
- Use **Terraform** for Infrastructure as Code.
- Implement CI/CD pipelines for automated deployment.

Service Discovery and Scaling

Service Discovery

- Kubernetes automatically discovers services.
- Internal DNS enables communication between services.
- Load balancers direct traffic to healthy containers.

Scaling

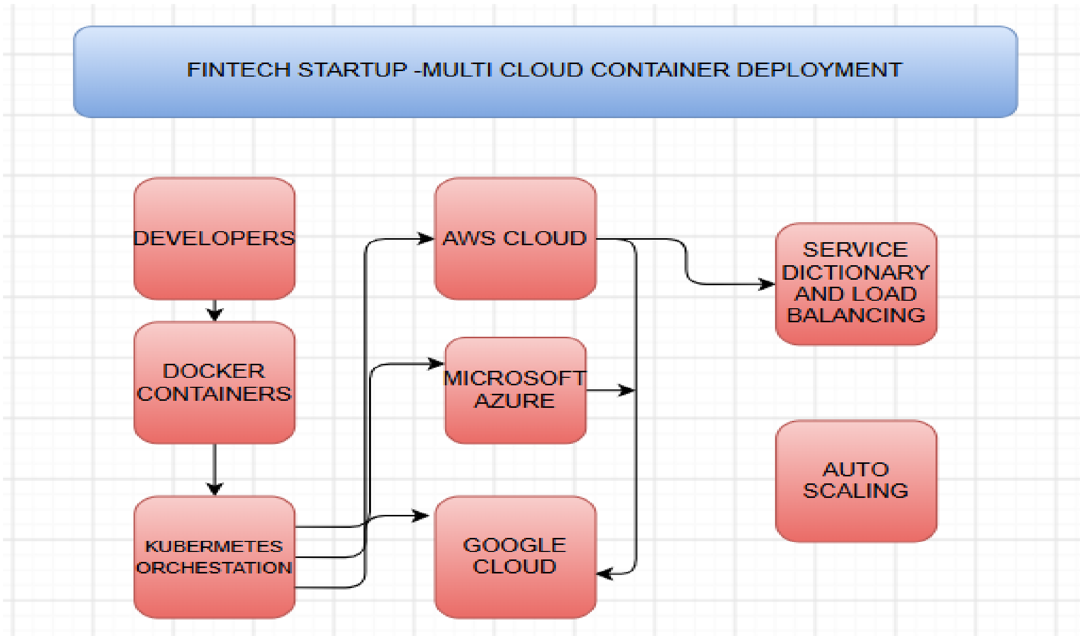
- Horizontal Pod Autoscaler increases or decreases pods automatically.
- Cluster Autoscaler adjusts worker nodes.
- Scaling is based on CPU and memory utilization.

Benefits of Multi-Cloud

- High availability
- Vendor independence
- Better disaster recovery
- Improved performance
- Business continuity

Risks

- Complex management
- Higher networking costs
- Data synchronization issues
- Security policy differences
- Monitoring complexity



3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how Recovery Time Objective (RTO) and Recovery Point Objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

Introduction

Data loss due to accidental deletion can disrupt learning services. Cloud-native backup and disaster recovery help restore services quickly and prevent permanent data loss.

Backup and Disaster Recovery Solution

- Enable automatic cloud backups.
- Store backups in multiple availability zones.
- Enable object versioning.
- Configure cross-region replication.
- Maintain disaster recovery sites.
- Perform regular backup verification.

Achieving RTO and RPO

Recovery Time Objective (RTO)

- Automated failover restores services quickly.
- Replicated infrastructure minimizes downtime.

Recovery Point Objective (RPO)

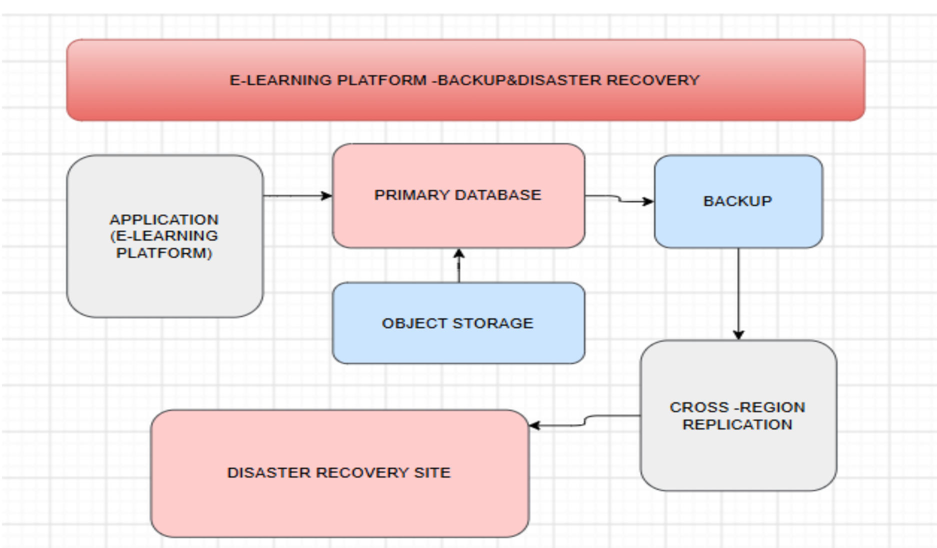
- Frequent backups reduce data loss.
- Continuous replication keeps backup copies updated.

Business Continuity Steps

1. Schedule automatic backups.
2. Enable cross-region replication.
3. Test backup restoration regularly.
4. Configure disaster recovery automation.
5. Monitor backup health continuously.
6. Maintain documented recovery procedures.

Advantages

- Fast recovery
- High data availability
- Reduced data loss
- Improved reliability
- Business continuity



4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

Introduction

Government agencies require secure storage of sensitive information while using public cloud resources for non-sensitive workloads. Hybrid cloud provides security, compliance, and scalability.

Hybrid Cloud Architecture

- Keep sensitive applications on-premises.
- Deploy non-sensitive applications in the public cloud.
- Connect environments using VPN or dedicated private links.
- Use API Gateway for secure communication.
- Monitor both environments through centralized tools.

Identity Management

- Implement Single Sign-On (SSO).
- Enable Multi-Factor Authentication (MFA).
- Use Role-Based Access Control (RBAC).
- Apply least-privilege access policies.

Encryption

- Encrypt stored data using AES-256.
- Protect data in transit using TLS/SSL.
- Manage encryption keys using Key Management Services (KMS).

Challenges

- Infrastructure complexity
- Compliance management
- Network latency
- Security integration
- Increased operational cost

Advantages

- Strong security
- Regulatory compliance
- Flexible deployment
- Better scalability
- Efficient resource utilization

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Introduction

High availability is essential for SaaS applications to ensure uninterrupted service. Multi-region deployment improves fault tolerance and disaster recovery.

Multi-Region Deployment Strategy

- Deploy applications in multiple cloud regions.
- Use Global Load Balancers.
- Configure Active-Active or Active-Passive architecture.
- Deploy replicated databases across regions.
- Synchronize application data continuously.

Data Replication and Failover

Data Replication

- Use synchronous replication for critical applications.
- Use asynchronous replication for geographically distant regions.
- Continuously synchronize database updates.

Failover

- Monitor regional health.
- Automatically redirect users to healthy regions.
- Restart services without manual intervention.

Monitoring and Alerting

- Monitor CPU, memory, storage, and network.

- Track application response time.
- Configure alerts for failures and threshold violations.
- Use centralized logging and monitoring dashboards.
- Perform regular health checks.

Benefits

- High availability
- Fault tolerance
- Business continuity
- Reduced downtime
- Better customer satisfaction